



# **NVIDIA Virtual GPU Software for Generic Linux with KVM Release Notes**

*Release 20.0-20.2*

**NVIDIA Corporation**

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# Chapter 1. Release Notes

These *Release Notes* summarize current status, information on validated platforms, and known issues with NVIDIA vGPU software and associated hardware on Linux with KVM.

## 1.1. NVIDIA vGPU Software Driver Versions

Each release in this release family of NVIDIA vGPU software includes a specific version of the NVIDIA Virtual GPU Manager, NVIDIA Windows driver, and NVIDIA Linux driver.

| NVIDIA vGPU Software Version | Soft-Manager Version | GPU | NVIDIA Driver Version | Windows | NVIDIA Linux Driver Version |
|------------------------------|----------------------|-----|-----------------------|---------|-----------------------------|
| 20.2                         | 595.91.04            |     | 596.86                |         | 595.91.07                   |
| 20.1                         | 595.71.03            |     | 596.36                |         | 595.71.05                   |
| 20.0                         | 595.58.02            |     | 595.97                |         | 595.58.03                   |

For details of which Linux with KVM releases are supported, see [Hypervisor Software Releases](#).

## 1.2. Compatibility Requirements for the NVIDIA vGPU Manager and Guest VM Driver

The releases of the NVIDIA vGPU Manager and guest VM drivers that you install must be compatible. If you install an incompatible guest VM driver release for the release of the vGPU Manager that you are using, the NVIDIA vGPU fails to load.

See [VM running an incompatible NVIDIA vGPU guest driver fails to initialize vGPU when booted](#).

### Note

You must use [NVIDIA License System](#) with every release in this release family of NVIDIA vGPU software. All releases in this release family of NVIDIA vGPU software are **incompatible** with all releases of the NVIDIA vGPU software license server.

## 1.2.1. Compatible NVIDIA vGPU Manager and Guest VM Driver Releases

The following combinations of NVIDIA vGPU Manager and guest VM driver releases are compatible with each other.

- ▶ NVIDIA vGPU Manager with guest VM drivers from the same release
- ▶ NVIDIA vGPU Manager with guest VM drivers from different releases within the same major release branch
- ▶ NVIDIA vGPU Manager from a later major release branch with guest VM drivers from the previous branch
- ▶ NVIDIA vGPU Manager from a later long-term support branch with guest VM drivers from the previous long-term support branch

### Note

When NVIDIA vGPU Manager is used with guest VM drivers from a different release within the same branch or from the previous branch, the combination supports **only** the features, hardware, and software (including guest OSes) that are supported on both releases.

For example, if vGPU Manager from release 20.2 is used with guest drivers from release 16.4, the combination does **not** support Windows Server 2019 because NVIDIA vGPU software release 20.2 does not support Windows Server 2019.

The following table lists the specific software releases that are compatible with the components in the NVIDIA vGPU software 20 major release branch.

| NVIDIA vGPU Software Component | Releases          | Compatible Software Releases                                                                                                                                                                             |
|--------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NVIDIA vGPU Manager            | 20.0 through 20.2 | <ul style="list-style-type: none"> <li>▶ Guest VM driver releases 20.0 through 20.2</li> <li>▶ All guest VM driver 19.&lt;x&gt; releases</li> <li>▶ All guest VM driver 19.&lt;x&gt; releases</li> </ul> |
| Guest VM drivers               | 20.0 through 20.2 | NVIDIA vGPU Manager releases 20.0 through 20.2                                                                                                                                                           |

## 1.2.2. Incompatible NVIDIA vGPU Manager and Guest VM Driver Releases

The following combinations of NVIDIA vGPU Manager and guest VM driver releases are incompatible with each other.

- ▶ NVIDIA vGPU Manager from a later major release branch with guest VM drivers from a production branch two or more major releases before the release of the vGPU Manager
- ▶ NVIDIA vGPU Manager from an earlier major release branch with guest VM drivers from a later branch

The following table lists the specific software releases that are incompatible with the components in the NVIDIA vGPU software 20 major release branch.

| NVIDIA vGPU Software Component | Releases          | Incompatible Software Releases                                                 |
|--------------------------------|-------------------|--------------------------------------------------------------------------------|
| NVIDIA vGPU Manager            | 20.0 through 20.2 | All guest VM driver releases 18.<x> and earlier, <b>except</b> 19.<x> releases |
| Guest VM drivers               | 20.0 through 20.2 | All NVIDIA vGPU Manager releases 19.<x> and earlier                            |

## 1.3. Updates in Release 20.2

### 1.3.1. New Features in Release 20.2

#### ► Security updates

When these updates are disclosed, full details are provided in a new public NVIDIA GPU Display Driver security bulletin, which is listed on the [NVIDIA Product Security](#) page. To be notified of the availability of this and future bulletins, use the link on the [NVIDIA Product Security](#) page to subscribe to receive email notifications of security bulletin updates.

#### ► Miscellaneous bug fixes as explained in [Issues Resolved in Release 20.2](#)

## 1.4. Updates in Release 20.1

### 1.4.1. New Features in Release 20.1

#### ► Security updates

When these updates are disclosed, full details are provided in a new public NVIDIA GPU Display Driver security bulletin, which is listed on the [NVIDIA Product Security](#) page. To be notified of the availability of this and future bulletins, use the link on the [NVIDIA Product Security](#) page to subscribe to receive email notifications of security bulletin updates.

#### ► Miscellaneous bug fixes as explained in [Issues Resolved in Release 20.1](#)

## 1.5. Updates in Release 20.0

### 1.5.1. New Features in Release 20.0

- Support for the fixed share scheduler for time-sliced vGPUs with different amounts of frame buffer on the same physical GPU
- Support for the Wayland display server protocol
- Miscellaneous bug fixes as explained in [Issues Resolved in Release 20.0](#)

## 1.5.2. Newly Supported Hardware and Software in Release 20.0

- ▶ Newly supported graphics cards:
  - ▶ NVIDIA RTX PRO 6000 Blackwell Server Edition liquid cooled
  - ▶ NVIDIA RTX PRO 4500 Blackwell Server Edition
- ▶ Newly supported generic Linux with KVM platforms:
  - ▶ Verge.io VergeOS

## 1.5.3. Feature Support Withdrawn in Release 20.0

- ▶ Disabling strict round robin policy is no longer supported.
- ▶ The following graphics cards are no longer supported:
  - ▶ Tesla M10
  - ▶ Tesla V100 SXM2
  - ▶ Tesla V100 SXM2 32GB
  - ▶ Tesla V100 PCIe
  - ▶ Tesla V100 PCIe 32GB
  - ▶ Tesla V100S PCIe 32GB
  - ▶ Tesla V100 FHHL
  - ▶ Quadro RTX 6000
  - ▶ Quadro RTX 6000 passive
  - ▶ Quadro RTX 8000
  - ▶ Quadro RTX 8000 passive

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## Chapter 2. Validated Platforms

This release family of NVIDIA vGPU software provides support for several NVIDIA GPUs on validated server hardware platforms, Linux with KVM hypervisor software versions, and guest operating systems. It also supports the version of NVIDIA CUDA Toolkit that is compatible with R595 drivers.

### 2.1. Supported NVIDIA GPUs and Validated Server Platforms

This release of NVIDIA vGPU software on Linux with KVM provides support for several NVIDIA GPUs running on validated server hardware platforms.

For information about the NVIDIA GPUs supported by your specific hypervisor and the validated server hardware platforms on which they run, consult the documentation from your hypervisor vendor.

#### 2.1.1. GPUs Based on the NVIDIA Blackwell Architecture

##### **Note**

GPUs based on the NVIDIA Blackwell architecture support SR-IOV.

GPUs based on the NVIDIA Blackwell architecture support manual placement of vGPUs on GPUs in equal-size mode.

| GPU                                                        | Mixed vGPU Configuration - Linux with KVM Releases |                              | Supported NVIDIA vGPU Software Products <sup>1, 2, 3</sup>                              |    | 1:1 MIG-Backed NVIDIA vGPU <sup>5</sup>                                                 | Time-Sliced, MIG-Backed NVIDIA vGPU                                                     | GPU Pass Through                                                         |
|------------------------------------------------------------|----------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|                                                            | Frame Buffer Size (Mixed-Size Mode)                | Series                       | Time-Sliced NVIDIA vGPU Single-Instance GPU <sup>4</sup>                                | on |                                                                                         |                                                                                         |                                                                          |
| NVIDIA RTX PRO 6000 Blackwell Server Edition               | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> |    | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |
| NVIDIA RTX PRO 6000 Blackwell Server Edition liquid cooled | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> |    | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |
| NVIDIA RTX PRO 4500 Blackwell Server Edition               | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> |    | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |

## 2.1.2. GPUs Based on the NVIDIA Ada Lovelace Architecture

### Note

GPUs based on the NVIDIA Ada Lovelace architecture support SR-IOV.

GPUs based on the NVIDIA Ada Lovelace architecture support manual placement of vGPUs on GPUs

<sup>1</sup> The supported products are as follows:

- ▶ vWS: NVIDIA RTX Virtual Workstation
- ▶ vPC: NVIDIA Virtual PC
- ▶ vApps: NVIDIA Virtual Applications

<sup>2</sup> N/A indicates that the deployment is not supported.

<sup>3</sup> vApps is supported only on Windows operating systems.

<sup>4</sup> A single-instance GPU is a GPU that is not partitioned into multiple GPU instances.

<sup>5</sup> A 1:1 MIG-backed NVIDIA vGPU is a MIG-backed vGPU that occupies an entire GPU instance.

in equal-size mode.

| GPU                      | Mixed vGPU Configuration - Linux with KVM Releases |                              | Supported NVIDIA vGPU Software Products <small>Page 6, 1, Page 6, 2, f</small>          |                                                                          | GPU Through | Pass |
|--------------------------|----------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------|------|
|                          | Frame Size (Mixed-Size Mode)                       | Buffer Series                | NVIDIA vGPU                                                                             |                                                                          |             |      |
| NVIDIA L40S              | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA L40               | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA L20               | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA L20 liquid cooled | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA L4                | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA L2                | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA RTX 6000 Ada      | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA RTX 5880 Ada      | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |
| NVIDIA RTX 5000 Ada      | Consult vendor documentation                       | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |             |      |

## 2.1.3. GPUs Based on the NVIDIA Ampere Architecture

### Note

GPUs based on the NVIDIA Ampere architecture support SR-IOV.

The manual placement of vGPUs on GPUs in equal-size mode **is not** supported on GPUs based on the NVIDIA Ampere architecture.

| GPU                       | Mixed vGPU Configuration - Linux with KVM Releases |                              |                              |  | Supported NVIDIA vGPU Software Products <small>Page 6, 1, Page 6, 2, F</small> |  | GPU Through      | Pass |
|---------------------------|----------------------------------------------------|------------------------------|------------------------------|--|--------------------------------------------------------------------------------|--|------------------|------|
|                           | Frame Size (Mixed-Size Mode)                       | Buffer                       | Series                       |  | NVIDIA vGPU                                                                    |  |                  |      |
| NVIDIA A40 <sup>6</sup>   | Consult vendor documentation                       |                              | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |
| NVIDIA A16                | Consult vendor documentation                       |                              | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |
| NVIDIA A10                | Consult vendor documentation                       |                              | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |
| NVIDIA A2                 | Consult vendor documentation                       |                              | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |
| NVIDIA A6000 <sup>6</sup> | RTX                                                | Consult vendor documentation | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |
| NVIDIA A5500 <sup>6</sup> | RTX                                                | Consult vendor documentation | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |
| NVIDIA A5000 <sup>6</sup> | RTX                                                | Consult vendor documentation | Consult vendor documentation |  | ▶ vWS<br>▶ vPC<br>▶ vApps                                                      |  | ▶ vWS<br>▶ vApps |      |

<sup>6</sup> This GPU is supported only in displayless mode. In displayless mode, local physical display connectors are disabled.

## 2.1.4. GPUs Based on the NVIDIA Turing Architecture

### Note

SR-IOV and the manual placement of vGPUs on GPUs in equal-size mode are **not** supported on GPUs based on the NVIDIA Turing™ architecture.

| GPU      | Mixed Configuration -<br>vGPU Linux with KVM<br>Releases |                              | Supported NVIDIA Software Products<br><small>Page 6, 1, Page 6, 2, F</small>            |                                                                          | vGPU Prod-<br>ucts |  |
|----------|----------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------|--|
|          | Frame Size (Mixed-Size Mode)                             | Buffer Series                | NVIDIA vGPU                                                                             | GPU Through                                                              | Pass               |  |
| Tesla T4 | Consult vendor documentation                             | Consult vendor documentation | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vPC</li> <li>▶ vApps</li> </ul> | <ul style="list-style-type: none"> <li>▶ vWS</li> <li>▶ vApps</li> </ul> |                    |  |

## 2.1.5. Support for a Mixture of Time-Sliced vGPU Types on the Same GPU

Linux with KVM supports a mixture of different types of time-sliced vGPUs on the same physical GPU. Any combination of A-series, B-series, and Q-series vGPUs with any amount of frame buffer can reside on the same physical GPU simultaneously. The total amount of frame buffer allocated to the vGPUs on a physical GPU must not exceed the amount of frame buffer that the physical GPU has.

For example, the following combinations of vGPUs can reside on the same physical GPU simultaneously:

- ▶ A40-2B and A40-2Q
- ▶ A40-2Q and A40-4Q
- ▶ A40-2B and A40-4Q

By default, a GPU or GPU instance supports only vGPUs with the same amount of frame buffer and, therefore, is in equal-size mode. To support vGPUs with different amounts of frame buffer, the GPU or GPU instance must be put into mixed-size mode. In mixed-size mode, the maximum number of some types of vGPU allowed on a GPU or GPU instance is less than in equal-size mode. For more information, refer to [Virtual GPU Software User Guide](#).

## 2.2. Hypervisor Software Releases

NVIDIA vGPU software is supported on Linux with KVM platforms **only** by specific hypervisor software vendors. For information about which NVIDIA vGPU software releases and hypervisor software releases are supported, consult the documentation from your hypervisor vendor.

**Note**

For NVIDIA vGPU software support, contact your Linux with KVM hypervisor vendor. Throughout the support process, your Linux with KVM hypervisor vendor remains your primary point of contact. Your Linux with KVM hypervisor vendor will coordinate as necessary with NVIDIA to address any issues that are confirmed to be issues with NVIDIA vGPU software. Any software updates to address issues with NVIDIA vGPU software are distributed by your Linux with KVM vendor, **not** NVIDIA.

| Hypervisor Vendor        | Platform                         | Additional Information                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H3C                      | CAS                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Inspur                   | InCloud Sphere                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Nutanix                  | AHV                              | <p>Obtain the NVIDIA Virtual GPU Manager software directly from Nutanix through the <a href="#">My Nutanix</a> portal (My Nutanix account required).</p> <div> <b>Note</b> <p>If the NVIDIA vGPU software release that you need is not available from the My Nutanix portal, contact Nutanix.</p> </div> <p>Then follow the instructions on the My Nutanix portal to obtain the correct NVIDIA vGPU software graphics drivers from the NVIDIA Licensing Portal.</p> |
| Proxmox Server Solutions | Proxmox Virtual Environment (VE) | <a href="#">NVIDIA vGPU Software on Proxmox VE</a>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Red Hat                  | OpenStack Platform               | <a href="#">Configuring the Compute Service for Instance Creation</a>                                                                                                                                                                                                                                                                                                                                                                                               |
| Sangfor                  | aDesk                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| SUSE                     | Linux Server Enterprise          | <a href="#">SUSE Partner Software Catalog</a><br><a href="#">SUSE Linux Enterprise Server 15 - NVIDIA virtual GPU for KVM guests</a>                                                                                                                                                                                                                                                                                                                                |
| Verge.io                 | VergeOS                          | <a href="#">VergeOS Docs</a>                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 2.3. Guest OS Support

For information about Windows releases and Linux distributions supported as a guest OS, consult the documentation from your hypervisor vendor.

**Note**

Use only a guest OS release that is listed as supported by NVIDIA vGPU software with your virtualization software. To be listed as supported, a guest OS release must be supported not only by NVIDIA vGPU software, but also by your virtualization software. NVIDIA **cannot** support guest OS

releases that your virtualization software does not support.

NVIDIA vGPU software supports **only** 64-bit guest operating systems. No 32-bit guest operating systems are supported.

## 2.4. NVIDIA CUDA Toolkit Version Support

The releases in this release family of NVIDIA vGPU software support NVIDIA CUDA Toolkit 13.2.

To build a CUDA application, the system must have the NVIDIA CUDA Toolkit and the libraries required for linking. For details of the components of NVIDIA CUDA Toolkit, refer to [NVIDIA CUDA Toolkit 13.2 Release Notes](#).

To run a CUDA application, the system must have a CUDA-enabled GPU and an NVIDIA display driver that is compatible with the NVIDIA CUDA Toolkit release that was used to build the application. If the application relies on dynamic linking for libraries, the system must also have the correct version of these libraries.

For more information about NVIDIA CUDA Toolkit, refer to [CUDA Toolkit Documentation 13.2](#).

### Note

If you are using NVIDIA vGPU software with CUDA on Linux, avoid conflicting installation methods by installing CUDA from a distribution-independent runfile package. Do not install CUDA from a distribution-specific RPM or Deb package.

To ensure that the NVIDIA vGPU software graphics driver is not overwritten when CUDA is installed, deselect the CUDA driver when selecting the CUDA components to install.

For more information, see [NVIDIA CUDA Installation Guide for Linux](#).

## 2.5. Wayland Display Server Protocol Support

The Wayland display server protocol is supported with only a subset of supported GPUs, guest OS releases, and display resolutions.

### 2.5.1. Supported GPUs

All GPUs supported by NVIDIA vGPU software, **except** GPUs based on the NVIDIA Turing™ architecture.

### 2.5.2. Supported Guest OS Releases

Only Red Hat Enterprise Linux 10 guest OS releases supported by NVIDIA vGPU software.

- ▶ Ubuntu 26.04 LTS
- ▶ Ubuntu 24.04 LTS

### 2.5.3. Supported Display Resolutions

- ▶ 2560×1600
- ▶ 3840×2160
- ▶ 4096×2160
- ▶ 5120×2880
- ▶ 7680×4320

### 2.5.4. Limitations with Wayland Display Server Protocol Support

The following use cases are not supported:

- ▶ VDPAU
- ▶ Vulkan video playback
- ▶ Console VNC
- ▶ Frame-rate limiting (FRL)

### 2.5.5. Known Issues with Wayland Display Server Protocol Support

Refer to the online version of this document at [Known Issues](#).

## 2.6. Multiple vGPU Support

To support applications and workloads that are compute or graphics intensive, multiple vGPUs can be added to a single VM. The assignment of more than one vGPU to a VM is supported only on a subset of vGPUs and hypervisor software releases.

### 2.6.1. vGPUs that Support Multiple vGPUs Assigned to a VM

The supported vGPUs depend on the architecture of the GPU on which the vGPUs reside:

- ▶ For GPUs based on the NVIDIA Volta architecture and later GPU architectures, **all** Q-series vGPUs are supported.  
On GPUs that support the Multi-Instance GPU (MIG) feature, both time-sliced and MIG-backed vGPUs are supported.
- ▶ For GPUs based on the NVIDIA Pascal™ graphics architecture, only Q-series vGPUs that are allocated all of the physical GPU's frame buffer are supported.
- ▶ For GPUs based on the NVIDIA Maxwell™ graphics architecture, only Q-series vGPUs that are allocated all of the physical GPU's frame buffer are supported.

You can assign multiple vGPUs with differing amounts of frame buffer to a single VM, provided the board type and the series of all the vGPUs is the same. For example, you can assign an A40-48Q vGPU and an A40-16Q vGPU to the same VM. However, you cannot assign an A30-8Q vGPU and an A16-8Q vGPU to the same VM.

### 2.6.1.1 Multiple vGPU Support on the NVIDIA Blackwell Architecture

| Board                                                      | vGPU               |
|------------------------------------------------------------|--------------------|
| NVIDIA RTX PRO 6000 Blackwell Server Edition               |                    |
| NVIDIA RTX PRO 6000 Blackwell Server Edition liquid cooled | All Q-series vGPUs |
| NVIDIA RTX PRO 4500 Blackwell Server Edition               | All Q-series vGPUs |

### 2.6.1.2 Multiple vGPU Support on the NVIDIA Ada Lovelace Architecture

| Board                    | vGPU               |
|--------------------------|--------------------|
| NVIDIA L40S              | All Q-series vGPUs |
| NVIDIA L40               | All Q-series vGPUs |
| NVIDIA L20               |                    |
| NVIDIA L20 liquid cooled | All Q-series vGPUs |
| NVIDIA L4                | All Q-series vGPUs |
| NVIDIA L2                | All Q-series vGPUs |
| NVIDIA RTX 6000 Ada      | All Q-series vGPUs |
| NVIDIA RTX 5880 Ada      | All Q-series vGPUs |
| NVIDIA RTX 5000 Ada      | All Q-series vGPUs |

### 2.6.1.3 Multiple vGPU Support on the NVIDIA Ampere GPU Architecture

| Board            | vGPU                                     |
|------------------|------------------------------------------|
| NVIDIA A40       | All Q-series vGPUs <sup>7</sup>          |
| NVIDIA A16       | All Q-series vGPUs <sup>Page 16, 7</sup> |
| NVIDIA A10       | All Q-series vGPUs <sup>Page 16, 7</sup> |
| NVIDIA A2        | All Q-series vGPUs <sup>Page 16, 7</sup> |
| NVIDIA RTX A6000 | All Q-series vGPUs <sup>Page 16, 7</sup> |
| NVIDIA RTX A5500 | All Q-series vGPUs <sup>Page 16, 7</sup> |
| NVIDIA RTX A5000 | All Q-series vGPUs <sup>Page 16, 7</sup> |

#### 2.6.1.4 Multiple vGPU Support on the NVIDIA Turing GPU Architecture

| Board    | vGPU               |
|----------|--------------------|
| Tesla T4 | All Q-series vGPUs |

### 2.6.2. Maximum Number of vGPUs Supported per VM

NVIDIA vGPU software supports up to a maximum of 16 vGPUs per VM.

### 2.6.3. Hypervisor Releases that Support Multiple vGPUs Assigned to a VM

For information about which generic Linux with KVM hypervisor software releases support the assignment of more than one vGPU device to a VM, consult the documentation from your hypervisor vendor.

## 2.7. Peer-to-Peer CUDA Transfers over NVLink Support

Peer-to-peer CUDA transfers enable device memory between vGPUs on different GPUs that are assigned to the same VM to be accessed from within the CUDA kernels. NVLink is a high-bandwidth interconnect that enables fast communication between such vGPUs. Peer-to-Peer CUDA transfers over NVLink are supported only on a subset of vGPUs, Linux with KVM releases, and guest OS releases.

### 2.7.1. vGPUs that Support Peer-to-Peer CUDA Transfers

Only Q-series time-sliced vGPUs that are allocated all of the physical GPU's frame buffer on physical GPUs that support NVLink are supported.

#### 2.7.1.1 Peer-to-Peer CUDA Transfer Support on the NVIDIA Ampere GPU Architecture

| Board            | vGPU      |
|------------------|-----------|
| NVIDIA A40       | A40-48Q   |
| NVIDIA RTX A6000 | A6000-48Q |
| NVIDIA RTX A5500 | A5500-24Q |
| NVIDIA RTX A5000 | A5000-24Q |

---

<sup>7</sup> This type of vGPU cannot be assigned with other types of vGPU to the same VM.

### 2.7.1.2 Peer-to-Peer CUDA Transfer Support on the NVIDIA Turing GPU Architecture

| Board                   | vGPU         |
|-------------------------|--------------|
| Quadro RTX 6000         | RTX6000-24Q  |
| Quadro RTX 6000 passive | RTX6000P-24Q |
| Quadro RTX 8000         | RTX8000-48Q  |
| Quadro RTX 8000 passive | RTX8000P-48Q |

### 2.7.1.3 Peer-to-Peer CUDA Transfer Support on the NVIDIA Volta GPU Architecture

| Board                | vGPU       |
|----------------------|------------|
| Tesla V100 SXM2 32GB | V100DX-32Q |
| Tesla V100 SXM2      | V100X-16Q  |

## 2.7.2. Hypervisor Releases that Support Peer-to-Peer CUDA Transfers

Peer-to-Peer CUDA transfers over NVLink are supported on all hypervisor releases that support the assignment of more than one vGPU to a VM. For details, see [Multiple vGPU Support](#).

## 2.7.3. Guest OS Releases that Support Peer-to-Peer CUDA Transfers

Linux only. Peer-to-Peer CUDA transfers over NVLink are **not** supported on Windows.

## 2.7.4. Limitations on Support for Peer-to-Peer CUDA Transfers

- ▶ NVSwitch is not supported. Only direct connections are supported.
- ▶ Only time-sliced vGPUs are supported. MIG-backed vGPUs are **not** supported.
- ▶ PCIe is not supported.
- ▶ SLI is not supported.

## 2.8. Unified Memory Support

Unified memory is a single memory address space that is accessible from any CPU or GPU in a system. It creates a pool of managed memory that is shared between the CPU and GPU to provide a simple

way to allocate and access data that can be used by code running on any CPU or GPU in the system. Unified memory is supported only on a subset of vGPUs and guest OS releases.

**Note**

Unified memory is disabled by default. If used, you must enable unified memory individually for each vGPU that requires it by setting a vGPU plugin parameter. NVIDIA CUDA Toolkit profilers are supported and can be enabled on a VM for which unified memory is enabled.

## 2.8.1. vGPUs that Support Unified Memory

On GPUs that support the MIG feature and on which this feature is enabled, **only** Q-series MIG-backed vGPUs that occupy an entire GPU instance are supported. All other MIG-backed vGPUs are **not** supported.

On single-instance GPUs, only Q-series vGPUs that are allocated all of the physical GPU's frame buffer on physical GPUs that support unified memory are supported.

### 2.8.1.1 Unified Memory Support on the NVIDIA Blackwell GPU Architecture

| Board                                                            | vGPU                                 |
|------------------------------------------------------------------|--------------------------------------|
| NVIDIA RTX PRO 6000 Blackwell Server Edition                     |                                      |
| NVIDIA RTX PRO 6000 Blackwell Server Edition liquid cooled       | NVIDIA RTX PRO 6000 Blackwell DC-96Q |
| All Q-series MIG-backed vGPUs that occupy an entire GPU instance |                                      |
| NVIDIA RTX PRO 4500 Blackwell Server Edition                     | NVIDIA RTX PRO 4500 Blackwell DC-32Q |
| All Q-series MIG-backed vGPUs that occupy an entire GPU instance |                                      |

### 2.8.1.2 Unified Memory Support on the NVIDIA Ada Lovelace GPU Architecture

| Board                    | vGPU             |
|--------------------------|------------------|
| NVIDIA L40               | L40-48Q          |
| NVIDIA L40S              | L40S-48Q         |
| NVIDIA L20               |                  |
| NVIDIA L20 liquid cooled | L20-48Q          |
| NVIDIA L4                | L4-24Q           |
| NVIDIA L2                | L2-24Q           |
| NVIDIA RTX 6000 Ada      | RTX 6000 Ada-48Q |
| NVIDIA RTX 5880 Ada      | RTX 5880 Ada-48Q |
| NVIDIA RTX 5000 Ada      | RTX 5000 Ada-32Q |

### 2.8.1.3 Unified Memory Support on the NVIDIA Ampere GPU Architecture

| Board            | vGPU      |
|------------------|-----------|
| NVIDIA A40       | A40-48Q   |
| NVIDIA A16       | A16-16Q   |
| NVIDIA A10       | A10-24Q   |
| NVIDIA A2        | A2-16Q    |
| NVIDIA RTX A6000 | A6000-48Q |
| NVIDIA RTX A5500 | A5500-24Q |
| NVIDIA RTX A5000 | A5000-24Q |

## 2.8.2. Guest OS Releases that Support Unified Memory

Linux only. Unified memory is **not** supported on Windows.

## 2.8.3. Limitations on Support for Unified Memory

- Only time-sliced Q-series and C-series vGPUs that are allocated all of the physical GPU's frame buffer on physical GPUs that support unified memory are supported. Fractional time-sliced vGPUs are **not** supported.

## 2.9. NVIDIA Deep Learning Super Sampling (DLSS) Support

NVIDIA vGPU software supports NVIDIA DLSS on NVIDIA RTX Virtual Workstation.

**Supported DLSS versions:** 2.0. Version 1.0 is **not** supported.

**Supported GPUs:**

- ▶ NVIDIA L40
- ▶ NVIDIA L40S
- ▶ NVIDIA L20
- ▶ NVIDIA L20 liquid cooled
- ▶ NVIDIA L4
- ▶ NVIDIA L2
- ▶ NVIDIA RTX 6000 Ada
- ▶ NVIDIA RTX 5880 Ada
- ▶ NVIDIA RTX 5000 Ada
- ▶ NVIDIA A40
- ▶ NVIDIA A16
- ▶ NVIDIA A2
- ▶ NVIDIA A10
- ▶ NVIDIA RTX A6000
- ▶ NVIDIA RTX A5500
- ▶ NVIDIA RTX A5000
- ▶ NVIDIA RTX PRO 6000 Blackwell Server Edition
- ▶ NVIDIA RTX PRO 6000 Blackwell Server Edition liquid cooled
- ▶ NVIDIA RTX PRO 4500 Blackwell Server Edition
- ▶ Tesla T4

**Note**

NVIDIA graphics driver components that DLSS requires are installed only if a supported GPU is detected during installation of the driver. Therefore, if the creation of VM templates includes driver installation, the template should be created from a VM that is configured with a supported GPU while the driver is being installed.

**Supported applications:** only applications that use `nvngx_dlss.dll` version 2.0.18 or newer

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## Chapter 3. Known Product Limitations

Known product limitations for this release of NVIDIA vGPU software are described in the following sections.

### 3.1. Maximum Allowed Network Adapters on a VM

If a VM is configured with too many network adapters, the VM might fail to acquire or return a license. The exact number of network adapters allowed on a VM depends on your system. In testing, it was verified that a VM with **40** network adapters could successfully acquire a license.

If a VM fails to acquire or return a license because the VM is configured with too many network adapters, the following error message is written to the licensing event log file:

```
Failed to update local trusted store - Maximum buffer size exceeded
```

For information about how to recover from such a failure, refer to [Virtual GPU Software Client Licensing User Guide](#).

### 3.2. NVENC does not support resolutions greater than 4096×4096

#### 3.2.1. Description

The NVIDIA hardware-based H.264 video encoder (NVENC) does not support resolutions greater than 4096×4096. This restriction applies to all NVIDIA GPU architectures and is imposed by the GPU encoder hardware itself, not by NVIDIA vGPU software. The maximum supported resolution for each encoding scheme is listed in the documentation for [NVIDIA Video Codec SDK](#). This limitation affects any remoting tool where H.264 encoding is used with a resolution greater than 4096×4096. Most supported remoting tools fall back to software encoding in such scenarios.

#### 3.2.2. Workaround

Use H.265 encoding. H.265 is more efficient than H.264 encoding and has a maximum resolution of 8192×8192.

**Note**

Resolutions greater than 4096×4096 are supported only by the H.265 decoder that 64-bit client applications use. The H.265 decoder that 32-bit applications use supports a maximum resolution of 4096×4096.

### 3.3. NVIDIA vGPU for Compute is not supported on Linux with KVM

NVIDIA vGPU for Compute is not supported on Linux with KVM. C-series vGPU types are not available.

### 3.4. Nested Virtualization Is Not Supported by NVIDIA vGPU

In general, NVIDIA vGPU deployments do not support nested virtualization, that is, running a hypervisor in a guest VM. For example, enabling the Hyper-V role in a guest VM running the Windows Server OS is **not** supported because it entails enabling nested virtualization. Similarly, enabling Windows Hypervisor Platform is not supported because it requires the Hyper-V role to be enabled.

### 3.5. Issues occur when the channels allocated to a vGPU are exhausted

#### 3.5.1. Description

Issues occur when the channels allocated to a vGPU are exhausted and the guest VM to which the vGPU is assigned fails to allocate a channel to the vGPU. A physical GPU has a fixed number of channels and the number of channels allocated to each vGPU is inversely proportional to the maximum number of vGPUs allowed on the physical GPU.

When the channels allocated to a vGPU are exhausted and the guest VM fails to allocate a channel, the following errors are reported on the hypervisor host or in an NVIDIA bug report:

```
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0): Guest
→attempted to allocate channel above its max channel limit 0xfb
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0): VGPU
→message 6 failed, result code: 0x1a
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0):
→0xc1d004a1, 0xff0e0000, 0xff0400fb, 0xc36f,
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0): 0x1,
→ 0xff1fe314, 0xff1fe038, 0x100b6f000, 0x1000,
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0):
→0x80000000, 0xff0e0200, 0x0, 0x0, (Not logged),
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0): 0x1,
```

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```
→ 0x0
Jun 26 08:01:25 srvxen06f vgpu-3[14276]: error: vmiop_log: (0x0): , 0x0
```

### 3.5.2. Workaround

Use a vGPU type with more frame buffer, thereby reducing the maximum number of vGPUs allowed on the physical GPU. As a result, the number of channels allocated to each vGPU is increased.

## 3.6. Virtual GPU hot plugging is not supported

NVIDIA vGPU software does not support the addition of virtual function I/O (VFIO) mediated device (mdev) devices after the VM has been started by QEMU. All mdev devices must be added before the VM is started.

## 3.7. Total frame buffer for vGPUs is less than the total frame buffer on the physical GPU

Some of the physical GPU's frame buffer is used by the hypervisor on behalf of the VM for allocations that the guest OS would otherwise have made in its own frame buffer. The frame buffer used by the hypervisor is not available for vGPUs on the physical GPU. In NVIDIA vGPU deployments, frame buffer for the guest OS is reserved in advance, whereas in bare-metal deployments, frame buffer for the guest OS is reserved on the basis of the runtime needs of applications.

If error-correcting code (ECC) memory is enabled on a physical GPU that does not have HBM2 memory, the amount of frame buffer that is usable by vGPUs is further reduced. All types of vGPU are affected, not just vGPUs that support ECC memory.

On all GPUs that support ECC memory and, therefore, dynamic page retirement, additional frame buffer is allocated for dynamic page retirement. The amount that is allocated is inversely proportional to the maximum number of vGPUs per physical GPU. All GPUs that support ECC memory are affected, even GPUs that have HBM2 memory or for which ECC memory is disabled.

The approximate amount of frame buffer that NVIDIA vGPU software reserves can be calculated from the following formula:

$$\text{<max-reserved-fb>} = \text{<vgpu-profile-size-in-mb>} \div 16 + 16 + \text{<ecc-adjustments>} + \text{<page-retirement-allocation>} + \text{<compression-adjustment>}$$

#### **<max-reserved-fb>**

The maximum total amount of reserved frame buffer in Mbytes that is not available for vGPUs.

#### **<vgpu-profile-size-in-mb>**

The amount of frame buffer in Mbytes allocated to a single vGPU. This amount depends on the vGPU type. For example, for the T4-16Q vGPU type, <vgpu-profile-size-in-mb> is 16384.

#### **<ecc-adjustments>**

The amount of frame buffer in Mbytes that is not usable by vGPUs when ECC is enabled on a physical GPU that does not have HBM2 memory.

- ▶ If ECC is enabled on a physical GPU that does not have HBM2 memory `<ecc-adjustments>` is `<fb-without-ecc>/16`, which is equivalent to 64 Mbytes for every Gbyte of frame buffer assigned to the vGPU. `<fb-without-ecc>` is total amount of frame buffer with ECC disabled.
- ▶ If ECC is disabled or the GPU has HBM2 memory, `<ecc-adjustments>` is 0.

**<page-retirement-allocation>**

The amount of frame buffer in Mbytes that is reserved for dynamic page retirement.

- ▶ On GPUs based on the NVIDIA Maxwell GPU architecture, `<page-retirement-allocation> = 4 ÷ <max-vgpus-per-gpu>`.
- ▶ On GPUs based on NVIDIA GPU architectures **after** the Maxwell architecture, `<page-retirement-allocation> = 128 ÷ <max-vgpus-per-gpu>`

**<max-vgpus-per-gpu>**

The maximum number of vGPUs that can be created simultaneously on a physical GPU. This number varies according to the vGPU type. For example, for the T4-16Q vGPU type, `<max-vgpus-per-gpu>` is 1.

**<compression-adjustment>**

The amount of frame buffer in Mbytes that is reserved for the higher compression overhead in vGPU types with 12 Gbytes or more of frame buffer on GPUs based on the Turing architecture.

`<compression-adjustment>` depends on the vGPU type as shown in the following table.

| vGPU Type                  | Compression Adjustment (MB) |
|----------------------------|-----------------------------|
| T4-16Q<br>T4-16C<br>T4-16A | 28                          |

For all other vGPU types, `<compression-adjustment>` is 0.

**Note**

In VMs running Windows Server 2012 R2, which supports Windows Display Driver Model (WDDM) 1.<x>, an additional 48 Mbytes of frame buffer are reserved and not available for vGPUs.

## 3.8. Issues may occur with graphics-intensive OpenCL applications on vGPU types with limited frame buffer

### 3.8.1. Description

Issues may occur when graphics-intensive OpenCL applications are used with vGPU types that have limited frame buffer. These issues occur when the applications demand more frame buffer than is allocated to the vGPU.

For example, these issues may occur with the Adobe Photoshop and LuxMark LuxCoreRender applications:

- ▶ When the image resolution and size are changed in Adobe Photoshop, a program error may occur or Photoshop may display a message about a problem with the graphics hardware and a suggestion to disable OpenCL.
- ▶ When the LuxMark LuxCoreRender application is run, XID error 31 may occur.

### 3.8.2. Workaround

For graphics-intensive OpenCL applications, use a vGPU type with more frame buffer.

## 3.9. In pass through mode, all GPUs connected to each other through NVLink must be assigned to the same VM

### 3.9.1. Description

In pass through mode, all GPUs connected to each other through NVLink must be assigned to the same VM. If a subset of GPUs connected to each other through NVLink is passed through to a VM, unrecoverable error XID 74 occurs when the VM is booted. This error corrupts the NVLink state on the physical GPUs and, as a result, the NVLink bridge between the GPUs is unusable.

### 3.9.2. Workaround

Restore the NVLink state on the physical GPUs by resetting the GPUs or rebooting the hypervisor host.

## 3.10. VM running an incompatible NVIDIA vGPU guest driver fails to initialize vGPU when booted

### 3.10.1. Description

A VM running a version of the NVIDIA guest VM driver that is incompatible with the current release of Virtual GPU Manager will fail to initialize vGPU when booted on a Linux with KVM platform running that release of Virtual GPU Manager.

A guest VM driver is incompatible with the current release of Virtual GPU Manager in either of the following situations:

- ▶ The guest driver is from a release in a branch two or more major releases before the current release, for example release 9.4.

In this situation, the Linux with KVM VM's `/var/log/messages` log file reports the following error:

```
vmiop_log: (0x0): Incompatible Guest/Host drivers: Guest VGX version is  
→older than the minimum version supported by the Host. Disabling vGPU.
```

- The guest driver is from a later release than the Virtual GPU Manager.

In this situation, the Linux with KVM VM's `/var/log/messages` log file reports the following error:

```
vmiop_log: (0x0): Incompatible Guest/Host drivers: Guest VGX version is  
→newer than the maximum version supported by the Host. Disabling vGPU.
```

In either situation, the VM boots in standard VGA mode with reduced resolution and color depth. The NVIDIA virtual GPU is present in Windows Device Manager but displays a warning sign, and the following device status:

```
Windows has stopped this device because it has reported problems. (Code 43)
```

## 3.10.2. Resolution

Install a release of the NVIDIA guest VM driver that is compatible with current release of Virtual GPU Manager.

## 3.11. Single vGPU benchmark scores are lower than pass-through GPU

### 3.11.1. Description

A single vGPU configured on a physical GPU produces lower benchmark scores than the physical GPU run in pass-through mode.

Aside from performance differences that may be attributed to a vGPU's smaller frame buffer size, vGPU incorporates a performance balancing feature known as Frame Rate Limiter (FRL). On vGPUs that use the best-effort scheduler, FRL is enabled. On vGPUs that use the fixed share or equal share scheduler, FRL is disabled.

FRL is used to ensure balanced performance across multiple vGPUs that are resident on the same physical GPU. The FRL setting is designed to give good interactive remote graphics experience but may reduce scores in benchmarks that depend on measuring frame rendering rates, as compared to the same benchmarks running on a pass-through GPU.

### 3.11.2. Resolution

FRL is controlled by an internal vGPU setting. On vGPUs that use the best-effort scheduler, NVIDIA does not validate vGPU with FRL disabled, but for validation of benchmark performance, FRL can be temporarily disabled by setting `frame_rate_limiter=0` in the vGPU configuration file.

```
# echo "frame_rate_limiter=0" > /sys/bus/mdev/devices/<vgpu-id>/nvidia/vgpu_
↳ params
```

For example:

```
# echo "frame_rate_limiter=0" > /sys/bus/mdev/devices/aa618089-8b16-4d01-a136-
↳ 25a0f3c73123/nvidia/vgpu_params
```

The setting takes effect the next time any VM using the given vGPU type is started.

With this setting in place, the VM's vGPU will run without any frame rate limit.

The FRL can be reverted back to its default setting as follows:

1. Clear all parameter settings in the vGPU configuration file.

```
# echo " " > /sys/bus/mdev/devices/<vgpu-id>/nvidia/vgpu_params
```

#### **Note**

You cannot clear specific parameter settings. If your vGPU configuration file contains other parameter settings that you want to keep, you must reinstate them in the next step.

2. Set `frame_rate_limiter=1` in the vGPU configuration file.

```
# echo "frame_rate_limiter=1" > /sys/bus/mdev/devices/<vgpu-id>/nvidia/
↳ vgpu_params
```

If you need to reinstate other parameter settings, include them in the command to set `frame_rate_limiter=1`. For example:

```
# echo "frame_rate_limiter=1 disable_vnc=1" > /sys/bus/mdev/devices/
↳ aa618089-8b16-4d01-a136-25a0f3c73123/nvidia/vgpu_params
```

## 3.12. nvidia-smi fails to operate when all GPUs are assigned to GPU pass-through mode

### 3.12.1. Description

If all GPUs in the platform are assigned to VMs in pass-through mode, `nvidia-smi` will return an error:

```
[root@vgx-test ~]# nvidia-smi
Failed to initialize NVML: Unknown Error
```

This is because GPUs operating in pass-through mode are not visible to `nvidia-smi` and the NVIDIA kernel driver operating in the Linux with KVM host.

To confirm that all GPUs are operating in pass-through mode, confirm that the `vfio-pci` kernel driver is handling each device.

```
# lspci -s 05:00.0 -k
05:00.0 VGA compatible controller: NVIDIA Corporation GM204GL [Tesla M60]
→(rev a1)
Subsystem: NVIDIA Corporation Device 113a
Kernel driver in use: vfio-pci
```

### 3.12.2. Resolution

N/A

---

## Chapter 4. Resolved Issues

Only resolved issues that have been previously noted as known issues or had a noticeable user impact are listed. The summary and description for each resolved issue indicate the effect of the issue on NVIDIA vGPU software **before the issue was resolved**.

### 4.1. Issues Resolved in Release 20.2

No resolved issues are reported in this release for Linux with KVM.

### 4.2. Issues Resolved in Release 20.1

No resolved issues are reported in this release for Linux with KVM.

### 4.3. Issues Resolved in Release 20.0

No resolved issues are reported in this release for Linux with KVM.



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## Chapter 5. Known Issues

Refer to the separate [Known Issues](#) document.



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## Chapter 6. Notices

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